HW03 — GUI & Usability

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0. Deliverable map

Deliverable	Link
GUI checklist (Markdown)	`features/GUI/checklist.md`
GUI checklist (Excel, 69 items)	`features/GUI/checklist.xlsx`
Test summary	`features/GUI/test-summary.md`
AI initial set	`features/GUI/ai-initial-checklist.md`
Review additions + rationale	`features/GUI/review-additions.md`
Bug report	`features/GUI/bug-report.md`
Individual bug reports (26)	`features/GUI/bugs/`
GitHub issue drafts (26)	`features/GUI/issues/`
Screenshots (failed items only)	`features/GUI/screenshots/`
AI audit report	`reports/ai-audit.md`

Task 1 — GUI checklist

1.1 Scope and objectives

The checklist targets the authentication and profile area of the EShop web frontend:

FR	Screen	Why it is in scope
FR-01 Account registration	`/register`	The first screen a new user meets; many validation and feedback states.
FR-02 Login and account lockout	`/login`	The entry point to every authenticated flow; security-sensitive input handling.
FR-03 Forgot password and reset	`/forgot-password`	Two-step flow with OTP, a state transition and error recovery.
FR-04 Profile management	`/profile`	Mixed form + data table; update, empty, loading and error states.

The four interface aspects required by the homework are covered explicitly:

- **IA-01 General UI standards** — layout, typography, colour/contrast, responsiveness, RTL, dark mode, zoom.
- **IA-02 Forms** — input types, labels, required fields, validation, password rules, OTP constraints.
- **IA-03 Navigation** — header, links, SPA routing, tab order, focus, route guards, current-page state.
- **IA-04 Feedback / state** — loading, success, error, empty, destructive-action confirmation, OTP privacy.

Objective: produce a non-repetitive checklist of more than 40 executable items, execute every item against the SUT, and separate genuine defects from design gaps.

1.2 Method

The work followed the course's AI-first but human-reviewed process:

1. **Initial AI generation with the `ui-checklist-generator` skill.** The skill's eight categories (layout, typography, colour, interaction, responsiveness, accessibility, product states, polish) were expanded across the four screens. This produced **44 initial items** (preserved in `ai-initial-checklist.md`).
2. **Critical review of the AI output.** The initial set was compared with what can actually fail on these screens: computed contrast, DOM/ARIA facts, keyboard order, SPA routing, step focus, route guards, responsive data tables, destructive actions and error/empty state separation. The review added **25 items** (see `review-additions.md`). These additions were produced in the same AI-assisted review pass and are still **pending the student's accept/reject decision**; the AI audit records them as agent output, not as human-authored work.
3. **Execution.** Every item was executed against a locally running SUT. Functional behaviour was driven with Playwright 1.65 (Chromium 158); DOM/ARIA facts were read from the live page; accessibility facts came from axe-core 4.x. Failed items were screenshotted with an annotation banner.
4. **Bug triage.** Failed items were grouped by root cause. A failed item is not automatically a bug: missing confirm-password fields and missing step indicators are recorded as design gaps, not defects. The reproducible defects became 26 bug reports.

1.3 Checklist design

69 items were designed and executed: 44 from the AI initial pass and 25 added during review. Every item belongs to exactly one IA and one FR/screen.

Interface aspect	Items	Passed	Failed
IA-01 General UI standards	19	8	11
IA-02 Forms	16	4	12
IA-03 Navigation	17	9	8
IA-04 Feedback / state	17	4	13
Total	**69**	**25**	**44**

Coverage is deliberately spread over the four screens rather than concentrated on one:

Screen	Items	Examples of what was checked
Login	17	heading, tab title, input types, label association, tab order, error/loading, authenticated redirect
Register	16	email format, duplicate email, password-rule consistency, confirm password, contrast, success feedback, raw server error
Forgot Password	16	two-step flow, OTP exposure, OTP input constraints, focus after transition, step indicator, button hierarchy, invalid-email alert
Profile	17	heading hierarchy, label association, phone validation, mobile table, order loading/error/empty, cancel confirmation, logout redirect
All auth screens	3	RTL layout, dark-mode legibility, 200% zoom

The complete table with the result and failure reason for every item is in `checklist.md` and `checklist.xlsx`.

1.4 Critical review: what the AI missed and why

The 25 review additions are listed with a per-item rationale in `review-additions.md`. The reasons fall into three groups.

A. Prompt quality

The first prompt asked for a GUI checklist for four screens but did not force the AI to produce an *executable oracle* per item. The AI therefore wrote broad statements such as “sufficient contrast” and “mobile usable” without a way to decide pass/fail. The review converted them into measurable checks (computed ratio ≥ 4.5:1, `scrollWidth = clientWidth` at 360px, `activeElement` after a step transition).

B. Model limitations

- **No execution.** The AI could not run the SPA, so it could not see that the forgot-password link causes a full reload, that `tabindex="1"` puts the submit button first, or that a 500 response is rendered as an empty order list.
- **Accessibility by appearance.** The AI treated visible `<label>` text as a label. It did not check `for` / `id`, `aria-label`, `aria-live`, `aria-invalid` or heading levels.
- **No computed contrast.** The AI said “check contrast” but did not calculate it. axe-core found 2.53:1, 3.76:1 and 3.92:1 failures.
- **Default assumptions.** The model assumed LTR, light mode and 100% zoom. RTL, dark mode and 200% reflow only appeared after an explicit review.
- **Feedback vs privacy.** The AI read “Mã OTP của bạn là: …” as helpful feedback; it did not flag it as a password-reset takeover vector.

C. Interface characteristics

Some defects are only visible in this particular SUT:

- The register/reset regex requires a whitespace character while the copy promises a special character — a contradiction that only appears when the regex and the copy are read together.

- The profile phone regex `/^[1-9][0-9]{8,9}$/` rejects every normal Vietnamese number starting with `0`, including the placeholder `0912345678`.
- The two-step reset flow keeps both steps in one component, so focus management and step indication are easy to miss.
- The order-history table has five columns and no responsive wrapper, so the page overflows only when an order exists.

1.5 Execution

Item	Value
Backend	`http://localhost:3000` (`backend/server.js`, seeded SQLite)
Web frontend	`http://localhost:5173` (`frontend-web`, Vite)
Browser	Chromium 158 / Playwright 1.65
Desktop viewport	1280×900
Mobile viewport	360×800
200% zoom equivalent	640×800
Accessibility engine	axe-core 4.x
Test users	`hw03.ui.23120223@example.com`, plus isolated `dup…` and `silent…` accounts
Order fixtures	Injected with Playwright route interception; no real order rows were created or modified

Execution rules:

- **Passed** = the observed behaviour matches the item's oracle in the executed environment.
- **Failed** = the observed behaviour contradicts the oracle. The Notes column records the exact observation.
- Screenshots were captured **only for failed items**; each screenshot has a red banner with the checklist ID and the observed defect.
- Contrast verdicts come from axe-core's computed ratios, not from visual judgement.

1.6 Results and analysis

1.6.1 Headline result

25 of 69 items passed and 44 failed. The failure rate is high because the SUT is intentionally seeded with UI/UX defects, and because the checklist deliberately includes accessibility and state-edge cases that a happy-path smoke test would skip.

1.6.2 Systemic issues (not isolated bugs)

1. **Authentication screens are visually and semantically unfinished.** The Login heading says "Đăng Ký", the password is visible, the email field is `type="text"`, and every form label is unassociated. This is one root pattern (copy/paste auth form without a semantic pass), not four unrelated defects.
2. **Feedback is either absent or blocking.** Login shows a generic unannounced div; Forgot Password and Profile use native `alert()`. There is no shared inline message/`aria-live`

pattern.

3. **State handling is binary.** Register/Login have no loading state; Profile shows “no orders” while loading and after a 500 response. The UI cannot distinguish “not started”, “empty”, “failed” and “done”.
4. **Accessibility is not part of the definition of done.** axe-core reports critical label violations on all four screens, no page has an `h1`, contrast fails in several places, tab order is broken by `tabindex="1"`, and RTL/dark-mode/zoom were never considered.
5. **Destructive and security-sensitive actions lack safeguards.** The OTP is printed on screen; cancel-order has no confirmation and is offered for shipping orders; route guards are missing for `/profile` and `/login`.

1.6.3 Bugs by severity

Severity	Count	Examples
Blocker	2	OTP exposed on screen (BUG-GUI-24); valid Vietnamese phone rejected, blocking profile save (BUG-GUI-14)
Major	12	visible password, missing labels, email/duplicate validation, password-rule contradiction, mobile table overflow, tab order, route guards, raw server error, alert-only feedback, order loading/error confusion, cancel without confirmation
Minor	12	wrong Login heading, shared tab title, identical step-2 buttons, missing `h1`, RTL, no active nav, no loading state, silent registration success, focus loss, full reload, unconstrained OTP, missing confirm password
Total	**26**	See `bug-report.md`

Each bug has a report with steps, expected/actual, root cause, suggested fix and screenshot in `features/GUI/bugs/`. The GitHub issue bodies are in `features/GUI/issues/`.

1.7 Limitations and open items

- **GitHub Issues are not published yet.** This environment has no GitHub CLI or token. The 26 issue bodies are ready in `features/GUI/issues/`; after `gh auth login`, run `bash features/GUI/issues/publish.sh` from `homework/hw03/ncuong`. The report does not claim issue links that do not exist.
- **Task 1 was executed on one browser engine (Chromium).** Firefox, Safari and the Expo Go platform belong to Task 3.
- **The AI initial set is preserved as generated.** The student must still review the 69 items and the 26 bugs before submitting; the AI audit records what the agent produced and what it could not verify.
- **Timestamps in the AI audit are marked “Time not available in source”** because the conversation export does not contain them.

1.8 Evidence index

- 44 failed items → 40 annotated screenshots in `screenshots/`, with `manifest.json` mapping each image to checklist IDs.
- axe-core results and DOM facts are summarized in the Notes column; the raw JSON is in `screenshots/manifest.json` and the execution logs are reproducible from the scripts in

`tools/`.
- Excel workbook: `checklist.xlsx` (sheets: Checklist, Summary, Execution).

Task 2 — Usability evaluation

Status: not started. To be completed after Task 1 review: flow selection, task scenario, SUS/UEQ-S, 7 participants, pilot, sessions, analysis.

Task 3 — Cross-browser / cross-platform

Status: not started. To be completed after Task 1 review: Chrome, Firefox and Safari (or Android Chrome / Expo Go), with student ID and full name overlaid on every screenshot.

Agent Skills

The skills used for this task are in `skills/`. The GUI checklist workflow is documented in `skills/ui-checklist-generator/SKILL.md` and `skills/gui-usability-workflow/SKILL.md`. The demo video is not recorded yet.

AI audit and critique

- AI Audit Report: `reports/ai-audit.md`
- AI Critique: `reports/ai-critique.md` (draft — student review pending)

Self-assessment

The self-assessment table is in `README.md`. It is intentionally left blank until the student reviews the Task 1 deliverables.